check: $\rho = \rho_1$ $\quad q = c\rho_1(1-1) = 0$

1

9/13/24

$\rho \ll \rho_1$ $\quad q \approx \frac{\rho}{\rho_1} \ll 1$ $\quad q \approx c\rho(1-0)$

$$q = c\rho$$

$\frac{\text{cars}}{\text{time}}$ $\quad \frac{\text{cars}}{\text{Length}}$ $\quad [c] = \frac{\text{Length}}{\text{time}}$ (velocity)

eliminate $q$

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}\left[c\rho\left(1 - \frac{\rho}{\rho_1}\right)\right] = 0$$

Start 9-13

$$\frac{\partial \rho}{\partial t} + c\rho_x\left(1 - \frac{\rho}{\rho_1}\right) + c\rho\left(-\frac{\rho_x}{\rho_1}\right) = 0$$

To be continued...

$$\frac{\partial \rho}{\partial t} + c\rho_x\left(1 - \frac{\rho}{\rho_1} - \frac{\rho}{\rho_1}\right) = 0$$

2

$$\frac{\partial \rho}{\partial t} + c\left[1 - 2\frac{\rho}{\rho_1}\right]\rho_x = 0$$

Minimize # of constants by nondimensionalization

$d = \frac{\rho}{\rho_1}$ dimensionless density

$\rho = 0 \quad d = 0$

$\rho = \rho_1 \quad d = 1$

$\vdots$

$\partial d$

$$\boxed{\begin{aligned} &\frac{\partial d}{\partial t} + c\left[1 - 2d\right]\frac{\partial d}{\partial x} = 0 \\ &\text{I.C.} \quad d(x,0) = f(x) \end{aligned}}$$

model for traffic flow

Recall Conserv. law

$$a(z)\, z_x + z_y = 0 \qquad z(x,0) = f(x)$$

$z \to d$ $\qquad a(d) = c(1-2d)$

$x \to x$

$y \to t$ $\qquad f(x) = f(x)$

sol'n: $z = f(x - a(z)\, y)$

$$d = f(x - c.(1-2d)t)$$

$$d = f(x - ct(1-2d)) \qquad \text{implicit sol'n.}$$

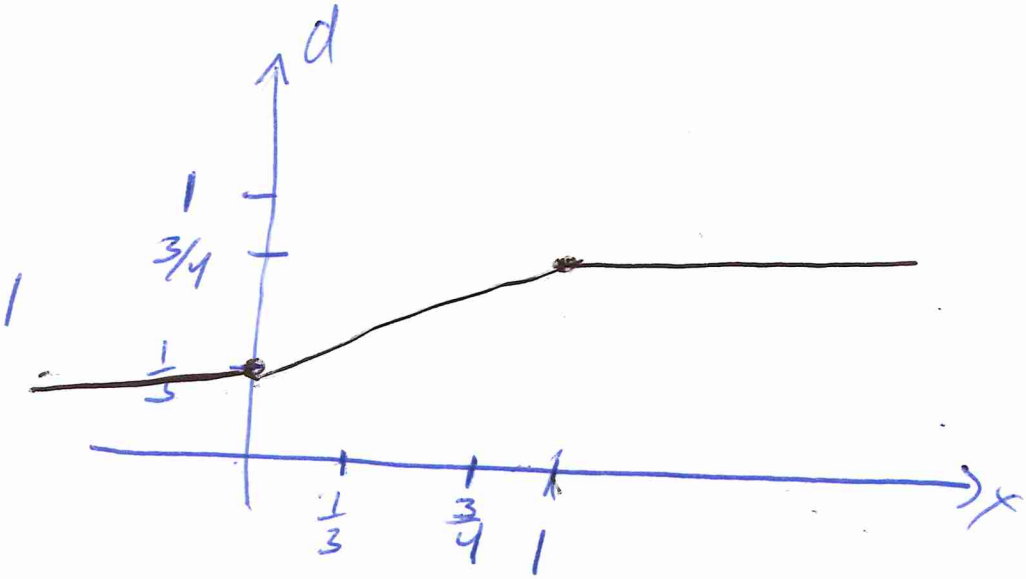
condition where sol'n exists:

$$1 - 2ct\, f'(x - ct(1-2d)) > 0.$$

example from Z+T text

$d(x,0) = f(x)$

$$f(x) = \begin{cases} \frac{1}{3} & x \le 0 \\ \frac{1}{3} + \frac{5}{12}x & 0 < x < 1 \\ 3/4 & x \ge 1 \end{cases}$$



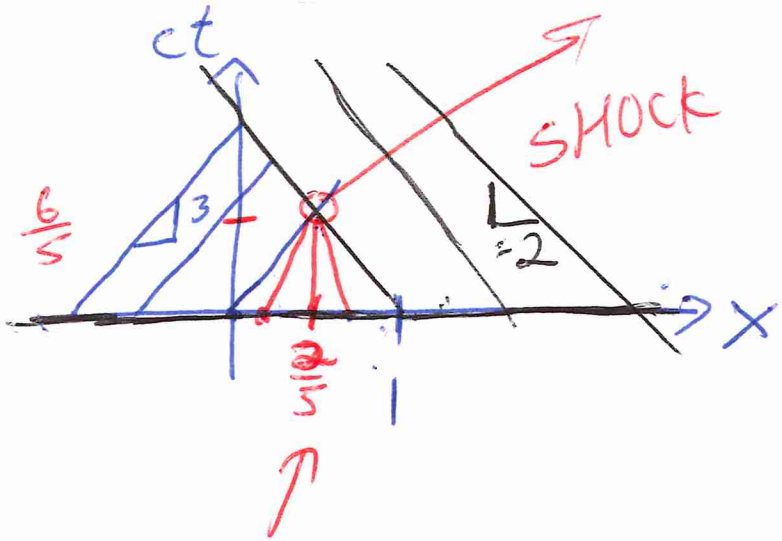

Note that $f(x) \notin C^1$, but if we just proceed, everything works (it turns out we just need $f(x)$ to be Lipschitz).

Sol'n $\quad d = f(x - ct(1-2d))$ exists

$1 - 2ct\, f'(x - ct(1-2d)) > 0$.

characteristics $\quad d = d_0$ on $x - ct(1-2d_0) = x_0$

$x_0 = \tau$




① when $x_0 \le 0$ $\quad d_0 = \frac{1}{3}$

$$ct = \frac{x - x_0}{1 - 2(\frac{1}{3})} = 3(x - x_0)$$

② when $x_0 \ge 1$ $\quad d_0 = 3/4$

$$ct = \frac{x - x_0}{1 - 2(\frac{3}{4})} = -2(x - x_0)$$

Where do these characteristics first intersect?

$ct = 3(x - 0)$ $\qquad ct = -2(x - 1)$

$3x = -2x + 2$

$5x = 2 \Rightarrow$

$x = 2/5$

$ct = 6/5$

③ $0 < x_0 < 1$ $\qquad f(x) = \frac{1}{3} + \frac{5}{12}x$ $\qquad d_0 = f(x_0) = \frac{1}{3} + \frac{5}{12}x_0$

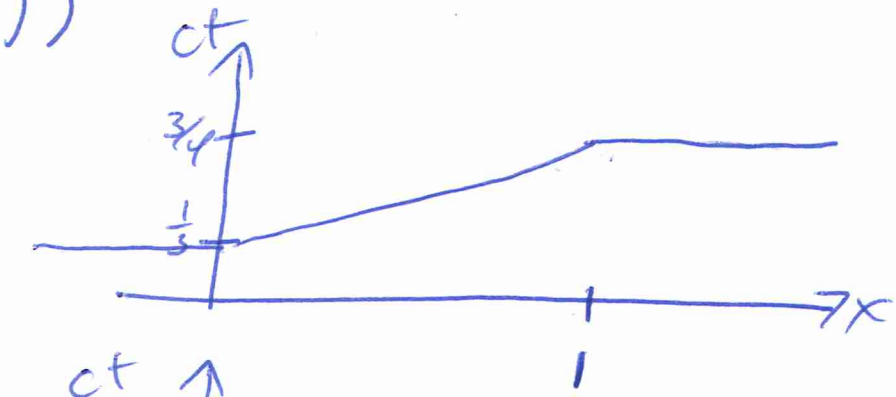
$$ct = \frac{x - x_0}{1 - 2d_0} = \frac{x - x_0}{1 - 2(\frac{1}{3} + \frac{5}{12}x_0)} = \dots = \frac{6(x - x_0)}{2 - 5x_0}$$

$x = 2/5$ $\qquad ct = \frac{6(\frac{2}{5} - x_0)}{2 - 5x_0} = \dots = \frac{6}{5}$ $\qquad$ all characteristics go through $(2/5, 6/5)$

Let us graph sol'ns for 4 different times

$$d = f(x - ct(1-2d))$$

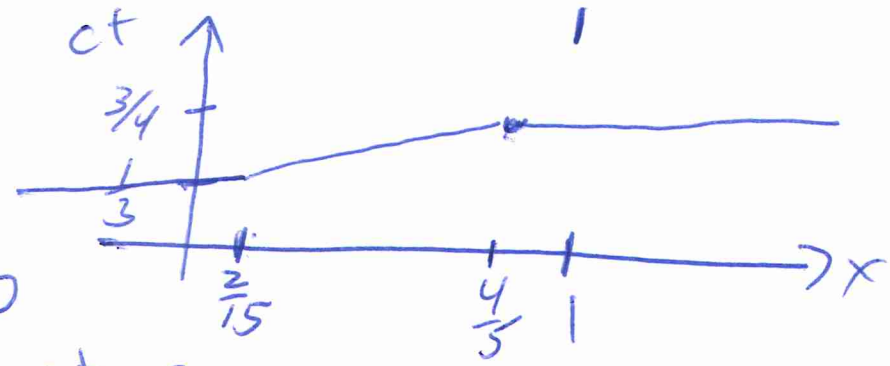
<u>$ct=0$</u> $\quad d = f(x)$




$ct=0$

<u>$ct = 2/5$</u>

$$d = f(x - 2/5(1-2d))$$
$$= f(x - \tfrac{2}{5} + \tfrac{4}{5}d)$$



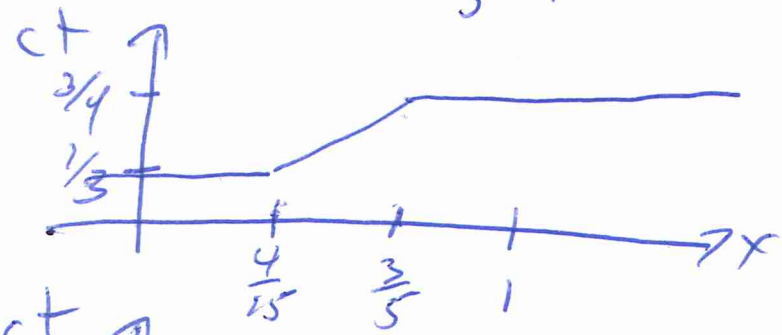

$ct=2/5$

$d = 1/3$ when $x - \frac{2}{5} + \frac{4}{5}(\frac{1}{3}) \le 0$

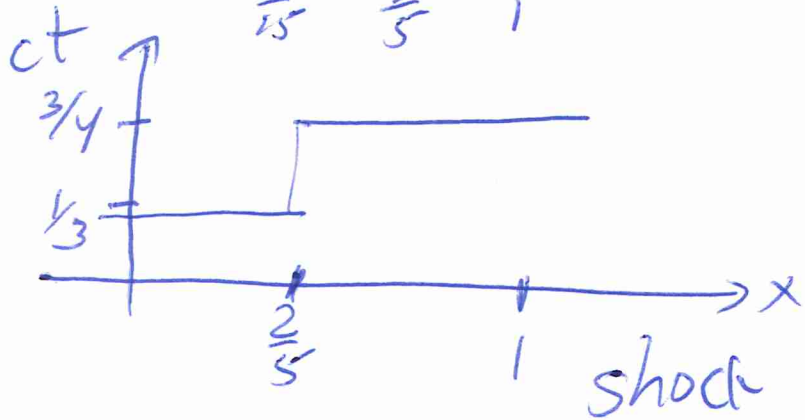
$$x \le \tfrac{2}{5} - \tfrac{4}{15} = 2/15$$

$d = 3/4 \quad x - \frac{2}{5} + \frac{4}{5}(3/4) \ge 1$

$$x \ge 4/5$$




$ct=4/5$




$ct = 6/5$